Project | Sustainability Impact Analysis for Intel

INTRODUCTION: As you learned listening in on the strategy meeting with Dr. Alvarez and Intel's Sustainability Team, Intel is committed to reducing its carbon footprint and improving the sustainability of its devices – not just during manufacturing, but throughout the entire lifecycle.

A key part of this effort is their repurposing programs, which play a central role in achieving these sustainability goals. Repurposing and recycling programs aim to reduce e-waste, energy consumption, and CO_2 emissions by extending the life of existing devices, and thus reducing the need for new device manufacturing. Like Michael Campbell said: the average household in the US has anywhere from 3–5 PCs devices, tablets, notebooks, desktops that are perfectly functional, but not being used!

One challenge Intel faces is determining which devices in its repurposing program should be prioritized for the maximum environmental benefit. That's where data analysis comes in! To help with this, Intel gathered data on each device repurposed or recycled in 2024.

Your task is to evaluate the effectiveness of Intel's current repurposing strategy and provide a data-driven recommendation to help guide the program's direction and optimize sustainability efforts.

HOW IT WORKS: Follow the prompts in the questions below to investigate the data. Post your answers in the provided boxes: the **yellow boxes** for the queries you write and **blue boxes** for your text-based analysis. Once you're done, you'll submit your ***completed*** .pdf file to HQ for feedback from The Accelerator Team.

SQL App: Here's the link to our specialized SQL app, where you'll write your SQL queries and interact with the data.

NOTE: The dataset you are working with is designed for the Global Career Accelerator to reflect the key characteristics and structure of Intel's real data, while protecting their confidentiality and proprietary information. Be aware that any conclusions or results derived from this dataset should be viewed as hypothetical and for illustrative purposes only.

— Data Set **Descriptions**

In this project you'll query 2 different datasets, `intel.device_data` and `intel.impact_data`, that you will join together for your analysis. Here you'll find the data dictionary for each dataset.

`intel.device_data`

- `device_id`: Unique identifier for each repurposed device
- `device_type`: Type of device, values are either "Laptop" or "Desktop"
- `model_year`: The year the device was manufactured (e.g., 2018, 2019, etc.)

`intel.impact_data`

- `impact_id`: Unique identifier for the repurposed device's impact record (e.g., "LP20NA141592")
- `device_id`: Unique identifier linking the impact record to a specific device in the `intel.device_data` table
- `usage_purpose`: The specific purpose for which the device is being repurposed, values are Education & Digital Literacy, Corporate & Enterprise, Government & Public Sector, Environmental Sustainability Programs, and Social Impact & Non-Profit
- `power_consumption`: Power consumption of the device in watts (W) when in use (e.g., 50W, 75W)
- `energy_savings_yr`: Estimated energy savings per device per year when repurposed compared to a new device, measured in kilowatt-hours (kWh)
- `co2_saved_kg_yr`: Estimated CO2 emissions saved per device per year from manufacturing a new device, measured in kilograms (kg).
- `recycling_rate`: The percentage of the device that is recyclable (e.g., 80%, 90%).
- `region`: The geographical region where the device was repurposed, values are "North America", "Europe", and "Asia"

— **Task 1:** Organizing and Understanding the Data

We'll start by **joining** the device data with the impact data, allowing for a comprehensive analysis of device types, model years, repurpose regions, and energy savings in one dataset.

A. Simply write a query that returns all of the columns from both tables, joining the two on the `device_id` column. Be sure to choose the appropriate join so that all relevant

data is included in your result. **Note:** your query will have more than 150,000 rows (the max display for SQLPad!)

(paste your query below 👇)

```
Select
  a.*,
  b.*
From
  intel.device_data as a FULL
  OUTER JOIN intel.impact_data as b
  on a.device_id = b.device_id
```

B. To your joined dataset, add a new column called `device_age` calculated by subtracting the model_year from 2024. Paste your query below and double check that the values in your new column make sense. For example, a 2019 device should be 5 years old.

(paste your query below 👇)

```
Select
  a.*,
  b.*,
  2024 - a.model_year as device_age
From
  intel.device_data as a FULL
  OUTER JOIN intel.impact_data as b
  on a.device_id = b.device_id
```

C. Order your joined data by model_year (oldest to newest). Do you notice more older (5+ years) or newer (under 5 years) devices being repurposed? What might that indicate?

(write your **answer** below 👇)

What i was able to notice from the older the devices is that yea many of there device ages was 5 or more, what this indicates is that the company has a sustainable lifecycle strategy. Instead of getting rid of old tech they found a way to get extra values out of it and givei it a second life.

D. Bucketing the `device_age` will allow us to analyze trends and patterns in energy savings and CO_2 reductions more effectively than using individual ages. Use a `CASE WHEN` clause to add one more column, called `device_age_bucket`, to your data, that is based on the device_age:

- `WHEN` the `device_age` is less than or equal to 3, `device_age_bucket` should be "newer"
- `WHEN` the `device_age` is greater than 3 but less than or equal to 6, `device_age_bucket` should be "mid-age"
- `WHEN` the `device_age` is greater than 6, `device_age_bucket` should be "older"

HINT: Instead of using e.g. `device_age <= 3`, you need to reference the calculation directly: `2024 - d.model_year <= 3`.

Double check that the values in your new column make sense! For example, a 2019 device should be characterized as "mid-age".

(paste your query below 👇)

```
Select
  a.*,
  b.*,
  CASE
    when (2024 - a.model_year) <= 3 then 'newer'
    when (2024 - a.model_year) > 3
    and (2024 - a.model_year) <= 6 then 'mid-age'
    else 'older'
  END as device_age_bucket,
```

```
  2024 - a.model_year as device_age
From
  intel.device_data as a FULL
  OUTER JOIN intel.impact_data as b on a.device_id =
b.device_id
```

— **Task 2:** Key Insights

Now it's time to analyze the overall impact of Intel's repurposing program. You will use your final query from **Task 1** together with the `WITH` keyword for the remainder of this Project as you aggregate and analyze the data you've organized and prepped. For a refresher, rewatch "🍿The `WITH` Keyword" in SkillBuilder 6.

A. What is the total number of devices Intel repurposed in 2024?

HINT: The dataset **is** representing all devices repurposed in 2024! You just need to COUNT all the rows in your joined data from Task 1!

(write your **answer** below 👇)

The total number of devices intel repurposed in 2024 is 601740 devices.

B. Write a query that returns the total number of devices repurposed, the average age of repurposed devices in 2024, the average estimated energy savings (kWh) from repurposed devices per year, and the total CO_2 emissions saved (in tons) from repurposed devices.

Note: CO_2 emissions are typically measured in tons. Since `CO₂_saved_kg_yr` is measured in kg, divide the `SUM(CO₂_saved_kg_yr)` by 1000 to report the total CO_2 emissions saved in tons.

(paste your query below 👇)

```
With based_data as (Select
  a.*,
  b.*,
  CASE
    when (2024 - a.model_year) <= 3 then 'newer'
    when (2024 - a.model_year) > 3
    and (2024 - a.model_year) <= 6 then 'mid-age'
    else 'older'
  END as device_age_bucket,
  2024 - a.model_year as device_age
From
  intel.device_data as a FULL
  OUTER JOIN intel.impact_data as b on a.device_id =
b.device_id)
Select
COUNT(*) as total_based_data,
ROUND(AVG(device_age), 2) as avg_device_age,
ROUND(AVG(energy_savings_yr), 2) AS
avg_energy_saving_kwh,
ROUND(SUM(energy_savings_yr) / 1000, 2) As
total_co2_saved_tons
from based_data
```

C. Now that you have calculated the average estimated energy savings (kWh) and CO_2 emissions saved (tons), use ChatGPT to help put these numbers into perspective.



Try this prompt: I found that each repurposed device saves approximately of XXX kWh of energy per year and Intel's repurposing program saved XXX tons of CO_2 emissions in one year. Help me understand the significance of these numbers. How would this compare to the energy consumption of a small city or the amount of CO_2 produced by cars? What is the environmental impact of these savings?

What comparisons did you find most impactful in terms of scale? Summarize how much energy and CO_2 emissions were saved and how it compares to something familiar, like powering households or reducing car emissions.

(write your **answer** below 👇)

What I found most impactful was the scale of the environmental benefits. The fact that these findings show such a large impact is impressive. Since the average vehicle emits about 4.6 tons of CO_2 per year, and the program prevented 15,490.05 tons of CO_2 emissions, it is equivalent to taking approximately 3,367 cars off the road for a year. This is a big deal because it helps conserve natural resources and decrease electronic waste. These comparisons helped me understand the impact the program has made because they connect the savings to things people can easily visualize. By repurposing devices instead of replacing them, the program helps reduce electronic waste and lower greenhouse gas emissions.

— **Task 3:** Identifying Trends & Maximizing Sustainability

By grouping our data in different ways, we can uncover patterns in energy savings and CO_2 reductions. These insights will help us determine which categories of devices contribute the most to sustainability efforts and where Intel should focus its repurposing strategy for maximum impact.

A. Write a query that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), grouped by device_type.

Note (again): You'll need to divide `AVG(CO₂_saved_kg_yr)` by 1000 to report the average CO_2 emissions saved in tons.

(paste your query below 👇)

```
With based_data as (Select
  a.*,
  b.*,
  CASE
```

```
    when (2024 - a.model_year) <= 3 then 'newer'
    when (2024 - a.model_year) > 3
    and (2024 - a.model_year) <= 6 then 'mid-age'
    else 'older'
  END as device_age_bucket,
  2024 - a.model_year as device_age
From
  intel.device_data as a FULL
  OUTER JOIN intel.impact_data as b on a.device_id =
b.device_id)
Select
device_type,

COUNT(*) as total_based_data,
ROUND(AVG(device_age), 2) as avg_device_age,
ROUND(AVG(energy_savings_yr), 2) as
avg_energy_saving_kwh,
ROUND(AVG(energy_savings_yr)/ 1000, 2) AS
avg_energy_saving_tons
from based_data
group by device_type
```

B. Based on the results, which device type contributes the most to energy savings and CO_2 reduction? Why might that be the case?

Hint: Don't forget you can use ChatGPT as your Teammate to help think through your response!

(write your **answer** below 👇)

Laptops contributes to the most energy saving and co2 reductions with over 408,064 compared to desktop with 193,676. This is because laptops are the most commonly used and replaced than desktops,

which is why laptop are leading to a larger numbers of devices available to repurpose.

C. Write a query that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), now grouped by `device_age_bucket`.

(paste your query below 👇)

```
With based_data AS (SELECT
  a.*,
  b.*,
  CASE
    when (2024 - a.model_year) <= 3 then 'newer'
    when (2024 - a.model_year) > 3
    and (2024 - a.model_year) <= 6 then 'mid-age'
    else 'older'
  END as device_age_bucket,
  2024 - a.model_year as device_age
FROM
  intel.device_data as a FULL
  OUTER JOIN intel.impact_data as b on a.device_id =
b.device_id)
SELECT
device_age_bucket,
COUNT(*) as total_based_data,
ROUND(AVG(device_age), 2) as avg_device_age,
ROUND(AVG(energy_savings_yr), 2) as
avg_energy_saving_kwh,
ROUND(AVG(energy_savings_yr)/ 1000, 2) AS
avg_energy_saving_tons
FROM based_data
GROUP BY device_age_bucket
```

D. Based on the result of your query, what do you notice about the relationship between device age and the number of devices repurposed versus the average energy saved?

(write your **answer** below 👇)

What i was able to notice about the about the relationship between the device age and the number of devices repurposed versus the average energy saved. Are the new devices were repurposed the most with lowest energy saved compared to the other. The mid-age devices were repurposed less frequently and and had a higher average energy saved compared to the newer devices. Also the older devices were repurposed the least amount yet they generated the highest energy savings .

E. Finally, write a query that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), now grouped by region.

(paste your query below 👇)

```
With based_data AS (SELECT
  a.*,
  b.*,
  CASE
    when (2024 - a.model_year) <= 3 then 'newer'
    when (2024 - a.model_year) > 3
    and (2024 - a.model_year) <= 6 then 'mid-age'
    else 'older'
  END as device_age_bucket,
  2024 - a.model_year as device_age
FROM
  intel.device_data as a FULL
  OUTER JOIN intel.impact_data as b on a.device_id =
b.device_id)
SELECT
region,
COUNT(*) as total_based_data,
```

```
ROUND(AVG(device_age), 2) as avg_device_age,
ROUND(AVG(energy_savings_yr), 2) as
avg_energy_saving_kwh,
ROUND(AVG(energy_savings_yr)/ 1000, 2) AS
avg_energy_saving_tons
FROM based_data
GROUP BY region
```

F. How does the carbon intensity of electricity in each region impact the total CO_2 savings from repurposed devices? Are there regions where repurposing leads to significantly higher environmental benefits? Why might that be?

(write your **answer** below 👇)

There isnt really that much of a significant difference in c02 between the regions Asia, North America, Europe since they are nearly identical amoung the three regions. North America produce the largest overall benefit due to the greater number of devices repurposed .

– **Task 4:** Data-Driven Recommendations

Using the findings from this analysis, we need to summarize key takeaways and develop actionable recommendations for Intel. Remember: the goal is to refine Intel's repurposing strategy to maximize energy savings and CO_2 reductions while ensuring the most effective use of resources.

A. Based on your analysis of the repurposed devices (including energy savings, CO_2 emissions, and device age), write **four** key takeaways in succinct sentences/bullets that summarize the most important patterns and insights from the data. These should be specific, concise, and focused on the implications of repurposing newer versus older devices.

(write your **answer** below 👇)

Older devices yield maximum per unit impact,

Newer dominate irepurposing volume but contribute less efficiently to sustiniable outcomes ,
Newer devices are repurposed the most, but provide the lowest energy and eo2 saving per device,
Mid-age devices sit in the middle, providing moderate saving and moderate purposing volume

B. Based on your four key takeaways and ChatGPT as your teammate, write a recommendation for Intel on how to improve the repurposing program. Your recommendation should include a clear action or strategy for Intel based on the data and a data-driven justification for why this approach would maximize energy savings and CO_2 reductions.

(write your **answer** below 👇)

Intel should protize older devices for immediate repurposing or refurbishment programs, since theory deliver the highest co2 reduction per unit. Because newere devices dominate total volume but having the lowest per divide environmental impact, this make them the alone less efficient for maximazing sustainability gains. Mid age devices provides a balanced contribution of both moderate volume and moderate efficienc, making them a strong leverage point. The total environmental benefit depends on both volume and efficiency, focusing only on high-volume newer devices or only high impact older devices would leave significant gains unrealized.

C. Briefly reflect on how ChatGPT's suggestions influenced your recommendation. Did it help you see something you hadn't considered? What parts of your recommendation were improved based on its response?

(write your **answer** below 👇)

Chat gpt suggestions helped me better structure my recommendation seperatingthe devices into clear segments older, mid age, and newer and how each of the groups contribute to overall environment impact. I

had alread notice that older devices had higher per unit c20 savings and new devices had a higher volume, but the feedback helped me connect those ideas into clearer strategy instead of treating them separately.

— **LevelUp:** Optimizing Repurposing Strategy for Maximum Impact

Now that you've gained insights into the energy savings and CO_2 reductions across different device types and regions, let's use this data to optimize Intel's repurposing strategy for maximum environmental benefit.

A. Add to your final query of Task 3 that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), grouped by region, **the percentage** of the total energy savings and CO_2 reductions contributed by each device type within each region.

HINT: To calculate the percentage of the total energy savings, use this formula:
`Total energy savings for the device type / Total energy savings for the region) * 100`
You'll use a similar one for the percentage of the total CO_2 reductions.



Try this prompt: What's the best way to calculate the percentage of CO_2 reductions contributed by each device type in each region?

(paste your query below 👇)

```
With based_data AS (SELECT
  a.*,
  b.*,
  CASE
    when (2024 - a.model_year) <= 3 then 'newer'
```

```
    when (2024 - a.model_year) > 3
    and (2024 - a.model_year) <= 6 then 'mid-age'
    else 'older'
  END as device_age_bucket,
  2024 - a.model_year as device_age
FROM
  intel.device_data as a FULL
  OUTER JOIN intel.impact_data as b on a.device_id =
b.device_id),


 regional as(
  SELECT
  region,
  device_type,
  COUNT(*) as device_in_categorey,
  SUM(energy_savings_yr) as  total_energy_saving_kwh,
  SUM(co2_saved_kg_yr) as total_co2_saving_kg
  FROM based_data
  GROUP BY
  region,
  device_type
  )
SELECT
region,
device_type,
device_in_categorey as total_device,
ROUND(total_energy_saving_kwh / device_in_categorey, 2)
as avg_energy_saving_kwh,
ROUND((total_co2_saving_kg/device_in_categorey )/ 1000,
2) as avg_co2_saved_ton,
ROUND((total_energy_saving_kwh/
SUM(total_energy_saving_kwh) OVER(partition by region )
)* 100, 2) as pct_of_region_energy_contributed,
ROUND((total_co2_saving_kg/ SUM(total_co2_saving_kg)
OVER(partition by region))* 100, 2)as
pct_of_region_co2_contributed
```

```
FROM regional
order BY
  region ASC, pct_of_region_co2_contributed desc
```

B. Based on the results of your query, analyze the data to answer:

- Which device types in which regions contribute the most energy savings and CO_2 reductions relative to their numbers?
- How can this analysis help Intel prioritize specific device types in certain regions to maximize environmental benefits?

(write your **answer** below 👇)

Based on the results, laptop contribute the most energy savings and co2 reductions in every region. Compared to the desktops which is on the lower side for each region. The main reason for this difference is much larger numbers of laptop being repurposed in each region although laptops and desktops are very similar on average energy saving there are more laptops being repurposed. Tis analysis can help intel priotize its sustainability efforts by focusing on increasing laptop repurposing programs, since they produce greater overall energy savingand co2 reductions across all regions.

C. In addition to focusing on sustainability, imagine Intel needs to optimize for cost-effectiveness in their repurposing program. How might you adjust your query to incorporate cost data (e.g., cost per repurposed device)? What strategies could Intel use to balance sustainability goals with cost constraints?

(write your **answer** below 👇)

I would add a colum tht contains the cost to repurpose each device. Intel should treat sustainability and cost as separate goal, find high enviormental impact per dollars spent, target regions with the best

returns on investment, set susstainabilty budgets and targets together, improve operational efficiency.

— Evaluation Rubric

Unlike your Milestones that were evaluated largely based on your effort, the evaluation of your Portfolio Project will follow traditional evaluation methods, with tasks assessed for correctness and assigned point values accordingly.

Partial credit will be given where parts of this task are correct, even if other parts are incorrect or incomplete.

Task title	Max points
Task 1: Organizing and Understanding the Data	40
Task 2: Key Insights	25
Task 3: Identifying Trends & Maximizing Sustainability	60
Task 4: Data-Driven Recommendations	75
TOTAL POINTS:	200
LevelUp	
Optimizing Repurposing Strategy for Maximum Impact	20